

作业 (4月29日)

p136 习题9-1

1 (1), 3, 4 (2)

pp150-151 习题9-2

1, 2 (1) (2) (5), 3 (1) (4)

4 (2), 5

10

14 (1) (4)

1. 设 $f(x,y)$ 为连续函数, 求 $\lim_{\rho \rightarrow 0} \frac{1}{\pi \rho^2} \iint_{x^2+y^2 \leq \rho^2} f(x,y) dx dy$ [习题 9-1 T3]

正确做法:

f 连续, 由二重积分中值定理: 存在 $P(x_0, y_0) \in G_\rho = \{(x,y) \mid x^2+y^2 \leq \rho^2\}$
最好写上 使得 $\frac{1}{\pi \rho^2} \iint_{G_\rho} f(x,y) d\sigma = \frac{1}{\pi \rho^2} \cdot f(x_0, y_0) \cdot S_{G_\rho} = f(x_0, y_0)$

由于 $\rho \rightarrow 0$, 故 $(x_0, y_0) \rightarrow (0,0)$, 再由 f 连续 $f(x_0, y_0) \rightarrow f(0,0)$

故原式极限为 $f(0,0)$

注: 用 ϵ - δ 语言按连续定义写也可

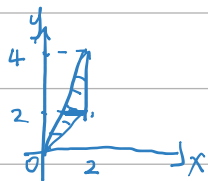
$\forall \epsilon > 0$, 存在 ρ_ϵ , 当 $x^2+y^2 \leq \rho_\epsilon^2$ 时 $|f(x,y) - f(0,0)| < \epsilon$

则 $|\frac{1}{\pi \rho^2} \iint_{x^2+y^2 \leq \rho^2} f(x,y) dx dy - f(0,0)| = |\frac{1}{\pi \rho^2} \iint_{x^2+y^2 \leq \rho^2} (f(x,y) - f(0,0)) dx dy| \leq \frac{1}{\pi \rho^2} \iint_{x^2+y^2 \leq \rho^2} |f(x,y) - f(0,0)| dx dy \leq \epsilon$

由 ϵ 任意性 $\lim_{\rho \rightarrow 0} \frac{1}{\pi \rho^2} \iint_{x^2+y^2 \leq \rho^2} f(x,y) dx dy = f(0,0)$

注: 不要都取极限 $\times$ 原式 $= \lim_{\rho \rightarrow 0} \frac{1}{\pi \rho^2} \iint_{x^2+y^2 \leq \rho^2} \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} f(x,y) dx dy = \lim_{\rho \rightarrow 0} \frac{1}{\pi \rho^2} \iint_{x^2+y^2 \leq \rho^2} f(0,0) dx dy = f(0,0)$

2. 交换积分顺序 $\int_0^2 dx \int_x^{2x} f(x,y) dy$ [习题 9-2 T3(1)]



$D = \{(x,y) \mid 0 \leq x \leq 2; x \leq y \leq 2x\} = \{(x,y) \mid 0 \leq y \leq 2, \frac{y}{2} \leq x \leq y\} \cup \{(x,y) \mid 2 \leq y \leq 4, \frac{y}{2} \leq x \leq 2\}$

原式 $= \iint_D f(x,y) dx dy = \int_0^2 dy \int_{\frac{y}{2}}^y f(x,y) dx + \int_2^4 dy \int_{\frac{y}{2}}^2 f(x,y) dx$ 分段

二重积分 = 重积分第一类曲线面积分 + 第二类曲线面积分

一. 重积分部分

1 (05-06) 计算 $\int_0^a dx \int_{-x}^{-a+\sqrt{a^2-x^2}} \frac{dy}{\sqrt{(x^2+y^2)(4a^2-x^2-y^2)}} \quad (a>0)$

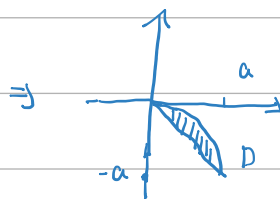
2 (05-06) 计算 $\int_{-1}^1 dx \int_0^{\sqrt{1-x^2}} \frac{1+y+xy^2}{1+x^2+y^2} dy$

3 (07-08) 设 D 是由曲线 $y=x^2$ 与直线 $y=x$ 所围成的平面区域

4 (07-08) 设 $D=\{(x,y) | 0 \leq x \leq 1, 0 \leq y \leq 1\}$, 计算 = 重积分 $\iint_D |x+y-1| dx dy$

1. $\int_0^a dx \int_{-x}^{-a+\sqrt{a^2-x^2}} \frac{dy}{\sqrt{(x^2+y^2)(4a^2-x^2-y^2)}} \quad (a>0)$

解: 积分区域 $\begin{cases} 0 \leq x \leq a \\ -x \leq y \leq -a+\sqrt{a^2-x^2} \end{cases} \Rightarrow \begin{cases} 0 \leq x \leq a \\ x+y \geq 0 \\ x^2+(y+a)^2 \leq a^2 \end{cases}$



极坐标换元 $x=r\cos\theta \quad y=r\sin\theta \quad \theta \in (-\pi, \pi)$

积分区域 $D = \{(r, \theta) | r \geq 0, -\pi \leq \theta \leq \pi; 0 \leq r \cos\theta \leq a; \cos\theta + \sin\theta \geq 0; 2a \sin\theta \leq -r\}$

$= \{(r, \theta) | -\frac{\pi}{4} \leq \theta \leq 0; 0 \leq r \leq \frac{a}{\cos\theta}, r \leq -2a \sin\theta\} \quad (-2a \sin\theta \leq \frac{a}{\cos\theta} \Leftrightarrow -\sin 2\theta \leq 1)$

$= \{(r, \theta) | -\frac{\pi}{4} \leq \theta \leq 0; 0 \leq r \leq -2a \sin\theta\}$

故原式: $\iint_D f(x,y) dx dy \stackrel{①}{=} \iint_D f^*(r, \theta) r dr d\theta = \int_{-\frac{\pi}{4}}^0 d\theta \int_0^{-2a \sin\theta} \frac{1}{r \sqrt{4a^2-r^2}} r dr$

$= \int_{-\frac{\pi}{4}}^0 d\theta \int_0^{-2a \sin\theta} \frac{1}{\sqrt{4a^2-r^2}} dr \stackrel{②}{=} \int_{-\frac{\pi}{4}}^0 d\theta [\arcsin \frac{r}{2a}]_0^{-2a \sin\theta} = \int_{-\frac{\pi}{4}}^0 (-\theta) d\theta = \frac{\pi^2}{32}$

① $\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{bmatrix} = r \quad ② (\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$

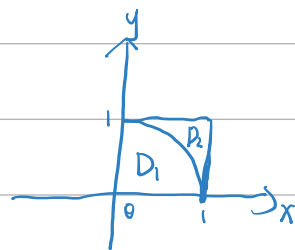
2. $\int_{-1}^1 dx \int_0^{\sqrt{1-x^2}} \frac{1+y+xy^2}{1+x^2+y^2} dy$

解: 积分区域 $D = \{(x,y) | -1 \leq x \leq 1; y \geq 0; x^2+y^2 \leq 1\} \stackrel{极}{=} \{(r, \theta) | 0 \leq r \leq 1; 0 \leq \theta \leq \pi\}$

原式: $\iint_D \frac{1+y+xy^2}{1+x^2+y^2} dy dx = \iint_D \frac{1+r\sin\theta+r^2\cos\theta\sin\theta}{1+r^2} r \cdot dr d\theta = \int_0^1 \frac{r}{1+r^2} dr \int_0^\pi (1+r\sin\theta+r^2\cos\theta\sin\theta) d\theta$

$= \int_0^1 \frac{r}{1+r^2} dr (\pi - r\cos\theta)_0^\pi + r^2 \int_0^\pi \sin\theta d\sin\theta = \int_0^1 (\pi+2r) \frac{r}{1+r^2} dr$

$= \int_0^1 \frac{\pi r}{1+r^2} dr + 2 \int_0^1 \frac{r^2}{1+r^2} dr = \frac{\pi}{2} \ln 2 + 2(1-\frac{\pi}{4})$



4. $D = \{(x,y) | 0 \leq x \leq 1, 0 \leq y \leq 1\}$ 计算 $\iint_D |x^2+y^2-1| dx dy$

解: [拆绝对值] $D = D_1 \cup D_2 = \{(x,y) | x \geq 0, y \geq 0, x^2+y^2 \leq 1\} \cup \{(x,y) | 0 \leq x \leq 1, 0 \leq y \leq 1, x^2+y^2 > 1\}$

记 $I_1 = \iint_{D_1} (x^2+y^2-1) d\sigma \quad I_2 = \iint_{D_2} (x^2+y^2-1) d\sigma \quad I = \iint_D (x^2+y^2-1) d\sigma \quad [I_1+I_2=I]$

原式: $I_2 - I_1 = (I - I_1) - I_1 = I - 2I_1$

I_2 不好直接去算

$I_1 = \int_0^{\frac{\pi}{2}} d\theta \int_0^1 (r^2-1) r dr = \frac{\pi}{2} (-\frac{1}{4}) = -\frac{\pi}{8}$

$\Rightarrow$ 原式: $\frac{\pi}{4} - \frac{1}{3}$

$I = \int_0^1 \int_0^1 (x^2+y^2-1) dx dy = -\frac{1}{3}$

二 三重积分 = 三重积分第一类曲面部分为平面区域积分 (续)

1 (14-15) 计算 $\iiint_{\Omega} \frac{dv}{\sqrt{x^2+y^2+z^2}}$, 其中 $\Omega = \{(x,y,z) | x^2+y^2+z^2 \leq 2z, z \geq 1, y \geq 0\}$.

2 (14-15) 设 Ω 是由 $z = \sqrt{x^2+y^2}$ 与平面 $z=1, z=2$ 所围成的区域, 求 $\iiint_{\Omega} \frac{e^z}{\sqrt{x^2+y^2+z^2}} dv$.

3 (13-14) 设 $\Omega = \{(x,y,z) | x^2+y^2 \leq 3z, 1 \leq z \leq 4\}$, 计算 $\iiint_{\Omega} \frac{dv}{\sqrt{x^2+y^2+z^2}}$.

4 (13-14) 设 $\Omega = \{(x,y,z) | \sqrt{x^2+y^2} \leq z \leq \sqrt{2-x^2-y^2}\}$, 求 $\iiint_{\Omega} (x+z) dv$.

5 (12-13) 求 $\iiint_{\Omega} |\sqrt{x^2+y^2+z^2}-1| dv$, 其中 $\Omega = \{(x,y,z) | \sqrt{x^2+y^2} \leq z \leq 1\}$.

6 (12-13) (1) 设 $f(y)$ 为连续函数, 证明 $\int_0^1 dz \int_0^z f(y) dy = \int_0^1 (1-y) f(y) dy$;

(2) 设 $\Omega = \{(x,y,z) | 0 \leq x \leq y, 0 \leq y \leq z, 0 \leq z \leq 1\}$, $f(x)$ 为连续函数, 证明

$$\iiint_{\Omega} f(x) dv = \frac{1}{2} \int_0^1 (x-1)^2 f(x) dx.$$

7 (10-11) 设 $f(t)$ 为连续函数, $F(u) = \int_0^u f(t) dt, F(1) = a$ (a 为常数), $\Omega = \{(x,y,z) | 0 \leq z \leq y, 0 \leq y \leq x, 0 \leq x \leq 1\}$,

计算三重积分 $\iiint_{\Omega} f(x)f(y)f(z) dv$.

1. $\Omega = \{(x,y,z) | x^2+y^2+z^2 \leq 2z, z \geq 1, y \geq 0\}$ 求 $\iiint_{\Omega} \frac{dv}{\sqrt{x^2+y^2+z^2}}$

解: 球坐标换元 $x = r \sin \varphi \cos \theta, y = r \sin \varphi \sin \theta, z = r \cos \varphi$ $0 \leq \varphi \leq \pi, 0 \leq \theta \leq 2\pi$

$$\Omega = \{(r, \varphi, \theta) | r^2 \leq 2r \cos \varphi, r \cos \varphi \geq 1, r \sin \varphi \sin \theta \geq 0, r \geq 0, 0 \leq \varphi \leq \pi, 0 \leq \theta \leq 2\pi\}$$

$$[\cos \varphi > 0, \sin \theta \geq 0, \frac{1}{\cos \varphi} \leq r \leq 2 \cos \varphi]$$

整理得 $\Omega = \{(r, \varphi, \theta) | 0 \leq \varphi \leq \frac{\pi}{4}, 0 \leq \theta \leq \pi, r \in [\frac{1}{\cos \varphi}, 2 \cos \varphi]\}$

$$\frac{\partial(x,y,z)}{\partial(r,\varphi,\theta)} = r^2 \sin \varphi$$

$$\text{原积分} = \iiint_{\Omega} \frac{1}{r} \cdot r^2 \sin \varphi dr d\varphi d\theta = \int_0^{\pi} d\theta \int_0^{\frac{\pi}{4}} \left(\int_{\frac{1}{\cos \varphi}}^{2 \cos \varphi} r dr \right) \sin \varphi d\varphi$$

$$= \pi \int_0^{\frac{\pi}{4}} (2 \cos^3 \varphi + \frac{1}{2 \cos \varphi}) \sin \varphi d\varphi = \pi \int_{\varphi=0}^{\frac{\pi}{4}} (2 \cos^2 \varphi + \frac{1}{2 \cos^2 \varphi}) d(-\cos \varphi)$$

$$\stackrel{t=\cos \varphi}{=} \pi \int_{\frac{\sqrt{2}}{2}}^1 2t^2 + \frac{1}{2t^2} dt = \pi \left(\frac{1}{6} + \frac{\pi}{3} \right)$$

(陈老师给的答案好像算错了)

2. Ω 是由 $z = \sqrt{x^2+y^2}, z=1, z=2$ 围成的区域. 求 $\iiint_{\Omega} \frac{e^z}{\sqrt{x^2+y^2+z^2}} dv$

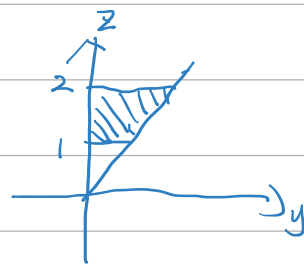
解: $\Omega = \{(x,y,z) | x^2+y^2 \leq z^2, 1 \leq z \leq 2\}$

柱坐标: $x = r \cos \theta, y = r \sin \theta$

$$\Omega = \{(r, \theta, z) | 0 \leq r \leq z, 1 \leq z \leq 2, 0 \leq \theta \leq 2\pi\}$$

$$\text{原式} = \iiint_{\Omega} \frac{e^z}{r} r dr d\theta dz = \int_0^{2\pi} d\theta \int_1^2 e^z \left(\int_0^z dr \right) dz = 2\pi \int_1^2 z e^z dz = 2\pi e^2$$

[事实上这里应当写成广义积分的形式. $z^2 = x^2+y^2 \leq z^2$ 避免分母出现 0]



3. $\Omega = \{(x,y,z) | x^2+y^2 \leq 3z, 1 \leq z \leq 4\}$ 求 $\iiint_{\Omega} \frac{1}{\sqrt{x^2+y^2+z^2}} dv$

柱坐标 $x = r \cos \theta, y = r \sin \theta, z = z$

$$\Omega = \{(x,y,z) | r^2 \leq 3z, 1 \leq z \leq 4, r \geq 0, 0 \leq \theta \leq 2\pi\}$$

$$\text{原式} = \iiint_{\Omega} \frac{1}{\sqrt{r^2+z^2}} r dr d\theta dz = \int_0^{2\pi} d\theta \int_1^4 \left(\int_0^{\sqrt{3z}} \frac{r}{\sqrt{r^2+z^2}} dr \right) dz$$

$$= 2\pi \int_1^4 \left(\sqrt{r^2+z^2} \Big|_{r=0}^{\sqrt{3z}} \right) dz = 2\pi \int_1^4 \sqrt{z} dz = \frac{28}{3}\pi$$

4. 求 $\iiint_{\Omega} (x+z) dv$ $\Omega = \{(x,y,z) | \sqrt{x^2+y^2} \leq z \leq \sqrt{2-x^2-y^2}\}$

解: 球坐标换元 $x = r \sin \varphi \cos \theta$ $y = r \sin \varphi \sin \theta$ $z = r \cos \varphi$ $r \geq 0$ $\varphi \in (0, \pi)$ $\theta \in (0, 2\pi)$

$$\Omega = \{(r, \varphi, \theta) | r \sin \varphi \leq r \cos \varphi \leq \sqrt{2-r^2} \sin \varphi\} = \{(r, \varphi, \theta) | \tan \varphi \leq 1, \cos \varphi \geq 0, 0 \leq r \leq \sqrt{2}\}$$

$$= \{(r, \varphi, \theta) | 0 \leq \varphi \leq \frac{\pi}{4}, 0 \leq \theta \leq 2\pi, 0 \leq r \leq \sqrt{2}\}$$

$$\text{原式} = \iiint_{\Omega} (r \sin \varphi \cos \theta + r \cos \varphi) r^2 \sin \varphi dr d\varphi d\theta = \int_0^{\sqrt{2}} r^3 dr \int_0^{\frac{\pi}{4}} d\varphi \int_0^{2\pi} d\theta (\sin^2 \varphi \cos \theta + \sin \varphi \cos \varphi)$$

$$(\int_0^{2\pi} \cos \theta d\theta = 0) = \int_0^{\sqrt{2}} r^3 dr \int_0^{2\pi} d\theta \cdot \int_0^{\frac{\pi}{4}} \sin \varphi \cos \varphi d\varphi = \frac{\pi}{2}$$

5. 求 $\iiint_{\Omega} |\sqrt{x^2+y^2+z^2}-1| dv$ $\Omega = \{(x,y,z) | \sqrt{x^2+y^2} \leq z \leq 1\}$

解: 球坐标 $x = r \sin \varphi \cos \theta$ $y = r \sin \varphi \sin \theta$ $z = r \cos \varphi$ $r \geq 0$ $\varphi \in (0, \pi)$ $\theta \in (0, 2\pi)$

$$\Omega = \{(r, \varphi, \theta) | r \sin \varphi \leq r \cos \varphi \leq 1, r \geq 0, \varphi \in (0, \pi), \theta \in (0, 2\pi)\}$$

$$= \{(r, \varphi, \theta) | \varphi \in (0, \frac{\pi}{4}), \theta \in (0, 2\pi), 0 \leq r \leq \frac{1}{\cos \varphi}\}$$

$$\text{原式} = \iiint_{\Omega} |r-1| r^2 \sin \varphi dr d\varphi d\theta = \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} \sin \varphi d\varphi \int_0^{\frac{1}{\cos \varphi}} r^2 |r-1| dr$$

$$\int_0^{\frac{1}{\cos \varphi}} r^2 |r-1| dr = \int_0^1 r^2 (1-r) dr + \int_1^{\frac{1}{\cos \varphi}} r^2 (r-1) dr = \frac{1}{6} + \frac{1}{4 \cos^3 \varphi} - \frac{1}{3 \cos^2 \varphi}$$

$$\text{原式} = 2\pi \int_0^{\frac{\pi}{4}} (\frac{1}{6} + \frac{1}{4 \cos^3 \varphi} - \frac{1}{3 \cos^2 \varphi}) \sin \varphi d\varphi$$

$$= -2\pi \int_{\varphi=0}^{\frac{\pi}{4}} (\frac{1}{6} + \frac{1}{4 \cos^3 \varphi} - \frac{1}{3 \cos^2 \varphi}) d \cos \varphi = \frac{\pi}{6} (2-1)$$

6. (1) 设 $F(y)$ 为连续函数, 证明 $\int_0^1 dz \int_0^z F(y) dy = \int_0^1 (1-y) F(y) dy$

(2) 设 $\Omega = \{(x,y,z) | 0 \leq x \leq y \leq z \leq 1\}$ $f(x)$ 为连续函数, 证明 $\iiint_{\Omega} f(x) dv = \frac{1}{2} \int_0^1 (x-1)^2 f(x) dx$

证明:

$$(1) \text{ LHS} = \iint_{0 \leq y \leq z \leq 1} F(y) dy dz = \int_0^1 F(y) dy \int_y^1 dz = \int_0^1 (1-y) F(y) dy = \text{RHS}$$

$$(2) \text{ LHS} = \iiint_{0 \leq x \leq y \leq z \leq 1} f(x) dx dy dz = \int_0^1 f(x) dx \int_x^1 dy \int_y^1 dz = \int_0^1 f(x) dx \int_x^1 (1-y) dy$$

$$= \int_0^1 f(x) dx (1-x + \frac{1}{2} x^2) = \frac{1}{2} \int_0^1 (x-1)^2 f(x) dx$$

7. 设 $f(t)$ 为连续函数, $F(x) = \int_0^x f(t) dt$, $F(1) = a$ (a 为常数) $\Omega = \{(x,y,z) | 0 \leq z \leq y \leq x \leq 1\}$

计算 $\iiint_{\Omega} f(x) f(y) f(z) dv$

注意到: $F(0) = \int_0^0 f(t) dt = 0$; $F(x) = f(x) \Rightarrow dF(x) = f(x) dx$

$$\text{解: 原式} = \int_0^1 f(x) dx \int_0^x f(y) dy \int_0^y f(z) dz = \int_0^1 f(x) dx \int_0^x f(y) F(y) dy = \int_0^1 f(x) dx \int_{y=0}^{y=x} F(y) dF(y)$$

$$= \int_0^1 f(x) dx (\frac{1}{2} F^2(x)) = \frac{1}{2} \int_0^1 F^2(x) f(x) dx = \frac{1}{2} \int_{x=0}^{x=1} F^2(x) dF(x) = \frac{1}{6} F^3(x) \Big|_{x=0}^{x=1} = \frac{a^3}{6}$$

